Better Baseline LIVE · Calculation service

Start with an assumption, not a prediction.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)
20
Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR
RR · ratio | ARR · pp
Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)
0.75
Decimal ratio; strictly >0 and <1.

Dropout in each arm (%)
0
One shared value for both arms. Valid: 0 to <100.

Target power (%)
80
>0 and <100; must exceed alpha.

Two-sided alpha (%)
5
>0 and <100.

Calculate & Lock Design

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

No live design yet

Preset inputs are not freshly calculated measurements. Calculate a design to unlock stress testing and simulation.

Awaiting calculation
Enrolled control: —
Enrolled treatment: —
Required evaluable counts: —
Analytic power: unavailable

Source: synthetic preset
Follow-up: {preset.follow_up_label}
No clinical source asserted.

Stress testing unavailable

LIVE ≠ clinical evidence

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

01 / INTERACTION · PRESET & RESET

Preset change/reset restores assumptions and seed/replicates, invalidates captured results and pending work. Identity/version/hash braces are binding placeholders, not fabricated identifiers. All preset options remain keyboard accessible. Power is conditional, not the probability a drug works.

Better Baseline

LIVE · Calculation service

An alternate hypothesis. The same workflow.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)
20
Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR
RR · ratio | ARR · pp
Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Absolute risk reduction (percentage points)
5
Enter 5 for five percentage points. Must be >0 and strictly smaller than control endpoint risk.

Dropout in each arm (%)
0
One shared value for both arms. Valid: 0 to <100.

Target power (%)
80
>0 and <100; must exceed alpha.

Two-sided alpha (%)
5
>0 and <100.

Recalculate & Lock Design

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

Stale · prior design retained

Planning assumptions changed. The prior enrolled counts remain locked until a new calculation succeeds. Dependent stress, charts and simulation are stale.

Previous lock · stale
Control enrolled: {previous.control_enrolled}
Treatment enrolled: {previous.treatment_enrolled}
Power: stale · not current

ARR = 5 percentage points
Source: manual user edit
Current inputs: unhashed until calculated

Stress testing unavailable

LIVE ≠ clinical evidence

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

02 / INTERACTION · EDITED ARR

Only ARR is shown; RR is hidden. Manual edits acquire manual-user provenance. Preserve the prior enrollment lock until successful recalculation. Block current downstream results and export. Brace values are service bindings, not live outputs.

03 / DESKTOP / WORKSPACE STATE

Better Baseline LIVE · Calculation service

Correct the assumption before calculating.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)
20
Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR
RR · ratio | ARR · pp
Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Absolute risk reduction (percentage points)
20
ARR must be >0 and below 20 percentage points. Focused first invalid field; value retained.

Dropout in each arm (%)
0
One shared value for both arms. Valid: 0 to <100.

Target power (%)
80
>0 and <100; must exceed alpha.

Two-sided alpha (%)
5
>0 and <100.

Calculate unavailable · fix ARR

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

Invalid planning inputs

Unable to calculate. ARR cannot equal or exceed baseline risk. Values are preserved; correct the ARR field and retry.

Previous lock · stale
Control enrolled: {previous.control_enrolled}
Treatment enrolled: {previous.treatment_enrolled}
Power: stale · not current

ARR = 20 percentage points
Source: manual user edit
Current inputs: invalid / unhashed

Stress testing unavailable

LIVE ≠ clinical evidence
Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

03 / INTERACTION · VALIDATION & FOCUS

On submit, announce the correction summary and focus ARR, the first invalid field. Set aria-invalid; connect the field error with aria-describedby. No request or lock replacement. Validate all ranges and target > alpha.

Better Baseline LIVE · Calculation service

Calculating your planned design...

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)
20
Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR
RR · ratio | ARR · pp
Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)
0.75
Decimal ratio; strictly >0 and <1.

Dropout in each arm (%)
0
One shared value for both arms. Valid: 0 to <100.

Target power (%)
80
>0 and <100; must exceed alpha.

Two-sided alpha (%)
5
>0 and <100.

Calculating... · submission disabled

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

Calculation in progress

Computing required enrollment. Entered values remain visible. No new lock exists until the response succeeds.

Loading · no values yet
Required evaluable counts: pending
Enrollment lock: pending
Analytic power: pending

Source: synthetic preset
Follow-up: {preset.follow_up_label}
No clinical source asserted.

Stress testing unavailable

LIVE ≠ clinical evidence

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

04 / INTERACTION · REQUEST IN FLIGHT

Disable duplicate submission; expose aria-busy and announce progress politely. If inputs change, invalidate pending work: late responses must not become current. Preserve inputs on failure. No invented percentage-complete indicator.

Better Baseline

LIVE · Calculation service

Enrollment is locked. Assumptions can move.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)
20
Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR
RR · ratio | ARR · pp
Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)
0.75
Decimal ratio; strictly >0 and <1.

Dropout in each arm (%)
0
One shared value for both arms. Valid: 0 to <100.

Target power (%)
80
>0 and <100; must exceed alpha.

Two-sided alpha (%)
5
>0 and <100.

Recalculate & Lock Design

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Design current

Enrollment locked
Control: {C enrolled}
Treatment: {T enrolled}
Before losses · immutable during stress testing

Metric	Control	Treatment
Required evaluable	{C required}	{T required}
Planned enrolled	{C enrolled}	{T enrolled}
Expected retained[1]	{C expected}	{T expected}
Endpoint risk	20%	{implied risk}
Nominal analytic power	{power at required n}	{same design power}

[1] Floored expected retained counts; not guaranteed. Sampled actual retained counts are only available after simulation.

Cohen’s-h / normal approximation
Model: {model_version}
Method: {method_version}
Inputs hash: {inputs_hash}
Scenario: {scenario_id}
Follow-up: {follow_up_label}
Generator: {generator}
Source/retrieval/verification: {returned metadata}

Calculation warnings

{service warnings, or explicit “None returned”}
Source: synthetic assumptions. No clinical source asserted.

Evaluate Stress Scenario

05 / INTERACTION · CURRENT SERVICE RESPONSE

LIVE success schema mockup: every brace binds to an actual service-return value, never a fabricated result. Current status requires matching inputs_hash. Successful recalculation replaces the lock and invalidates downstream captured results. Expand methods/warnings via keyboard-operable disclosure.

Better Baseline

Trial Planning Workspace / Stress-test assumptions. Keep enrollment fixed.

LIVE / SERVICE-BINDING PREVIEW

06 / CURRENT · Route / · Matching service bindings only; no fresh calculation. Chart and heatmap marks are illustrative scaffolds.

What happens when baseline risk falls?

↑ Planning assumptions

Baseline collapse · Synthetic assumptions · RR 0.75 · Target 80% · Two-sided α 5%

LOCKED ENROLLMENT

‹N_C› / ‹N_T›

Control / treatment participants
Enrolled before losses · immutable lock

Expected retained: ‹E_C› / ‹E_T›
Floored at selected dropout 0%

Stress edits never resize enrollment. Only a successful new design can replace this lock.

02 / STRESS SCENARIO

Change assumptions, not enrollment

Control endpoint risk · 10%

1% 99%

0.10
fraction (0.10 = 10%)

Symmetric dropout · 0%

0% 50%

0.00
fraction (0.10 = 10%)

Effect inherited from plan: RR 0.75. To change effect, edit planning inputs and recalculate.

Evaluate Stress Scenario | Load grid response from backend

Risk fraction 0.01–0.99; dropout 0–0.50, step 1pp. Debounce 350ms; explicit actions remain. Backend progress ‹phase›; on error ‹diagnostic› show Retry evaluation / Retry grid. Preserve enrollment; discard mismatched hashes.

03 / COMPARE

The same trial, different assumptions

Measure	Original plan	Selected stress
Control endpoint risk	20%	10%
Symmetric dropout	0% · planned	0% · selected
Frozen enrollment C / T	‹N_C› / ‹N_T› · fixed	‹N_C› / ‹N_T› · fixed
Approx. conditional power	‹original analytic power›	‹selected analytic power›
Power change	Reference	‹power change› percentage points

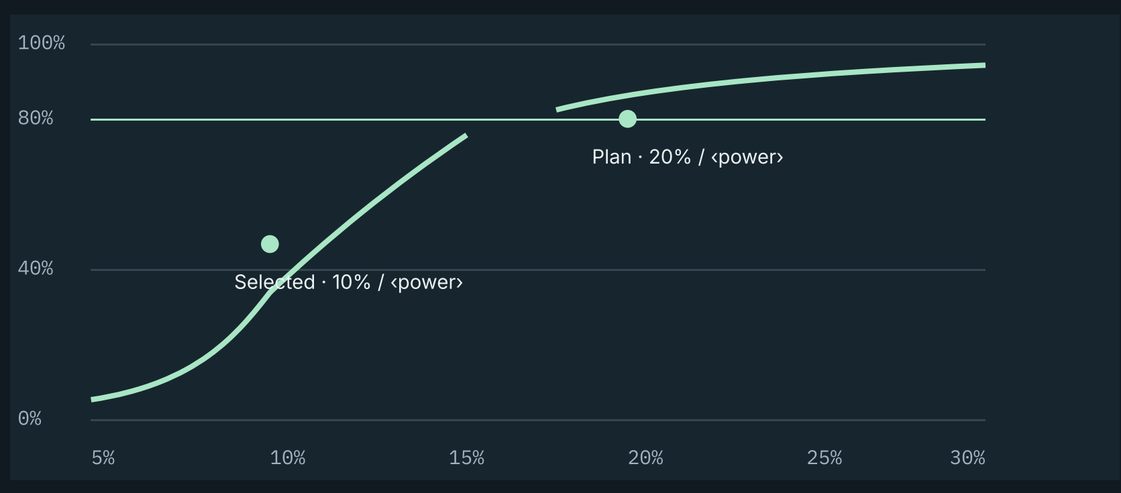
EXPECTED RETAINED C / T · Original ‹floor(N_C × (1−d_plan))› / ‹floor(N_T × (1−d_plan))› · Stress ‹floor(N_C × (1−d_stress))› / ‹floor(N_T × (1−d_stress))›

CURRENT · Target 80% · inherited RR 0.75 · Cohen's-h / normal approximation. Hash ‹matching hash› · version ‹method version› · warnings ‹diagnostics›. ARR plans retain the inherited ARR instead; never refit effect during stress.

04 / CURVE

Power at selected dropout

Fixed enrollment ‹N_C› / ‹N_T› · RR 0.75 · dropout exactly 0%
Analytic estimate, not Monte Carlo



Control baseline risk (%) → · Conditional power (%) ↑

Schematic trace only; not a computed dataset. Render backend valid series only, with gaps at nulls. Plan marker includes original dropout; if it differs, label separately from the curve at selected dropout.

Baseline	Power	Keyboard action
10%	‹analytic power›	Selected
20%	‹analytic power›	Select baseline
‹risk›	null · invalid	Unavailable

05 / SENSITIVITY

Find the target-power boundary

Control baseline × symmetric dropout · target 80%
Fixed enrollment and inherited effect · analytic estimates

Loss ↓	5%	10%	20%	30%
0%	‹power›	‹p› ✓	‹power›	‹power›
10%	‹power›	‹power›	‹power›	‹power›
20%	‹power›	‹power›	‹power›	‹power›
30%	null ×	‹power›	‹power›	‹power›

—— Target 80% boundary · ✓ selected · × invalid
Edge marks & tint are schematic only; bind returned boundary and power in production.

Boundary binds supplied segments, never across invalid regions. Axes ~5–30% risk / 0–30% loss; no domain editor. Selected scenarios outside axes still evaluate independently.

Download analytic grid CSV

Dropout ↓ / risk →	5%	10%	20%	30%
0%	‹power›	‹p› ✓	‹power›	‹power›
10%	‹power›	‹power›	‹power›	‹power›
20%	‹power›	‹power›	‹power›	‹power›
30%	null ×	‹power›	‹power›	‹power›

Each valid cell is a keyboard button: selects both risk and dropout, then refreshes. Invalid cells are disabled with explanation. CSV: fractions, nulls, invalid flags, selected markers, target, hashes, diagnostics.

Better Baseline

LIVE / SERVICE BINDINGS · 07 / STALE → LOADING · Route /

Enrollment stays locked. Results are updating.

STALE → LOADING · Evaluating current stress scenario…

Control risk changed from 10% to 12% (0.12). Dropout remains 0%. Inputs preserved; no enrollment recalculation.

LOCKED · ‹N_C› control / ‹N_T› treatment

Service bindings · retained = floor(N × (1−d)) / arm

02 / STRESS SCENARIO

Change assumptions, not enrollment

Control endpoint risk · 12%

1% 99%

0.12
fraction (0.10 = 10%)

Symmetric dropout · 0%

0% 50%

0.00
fraction (0.10 = 10%)

Effect inherited from plan: RR 0.75. To change effect, edit planning inputs and recalculate.

Evaluate Stress Scenario · pending

Load grid response from backend · pending

Risk 1–99%, dropout 0–50%, step 1pp; numeric fractions. STALE → LOADING after 350ms. Enrollment is unchanged.

Comparison · stale

Previous stress power ‹previous power› applies to 10% risk, not 12%. Δ pp unavailable. Original power ‹original power›; target 80%, inherited RR / ARR, method ‹version›, hash ‹previous hash›.

Curve and heatmap · loading

Selected dropout 0%. Previous trace/grid remain visibly stale; selected markers refresh only with matching results.

Simulation and export · unavailable

Scenario changed. Any previous simulation is stale; a complete matching simulation is required to export.

Evaluate Stress Scenario · pending

Load grid response · pending

350ms debounce; explicit Evaluate Stress Scenario and Load grid response use the same requests. Backend progress ‹phase / completed / total›. Failure ‹diagnostic› → Retry evaluation / Retry grid; preserve edits. Disable duplicates; discard mismatched hashes; polite live announcement.

Better Baseline

08 · Valid selection / simulation ready · Detail state of / — not a route

Simulate one selected scenario

LIVE / SERVICE-BINDING PREVIEW · VALID SELECTION

Selected grid point: control risk 10% · dropout 0%

FROZEN CONTEXT
Risk 10% (0.10) · dropout 0% (0.00)
Enrolled C / T: ‹N_C› / ‹N_T›
Retained: floor(N_C × (1−d)) / floor(N_T × (1−d))
Inherited RR 0.75 (or inherited ARR for ARR plan)
Target 80% · analytic power ‹matching power›
Cohen’s-h / normal approximation
Hash ‹matching inputs hash› · version ‹method version›
Warnings ‹diagnostics›

A valid grid cell updates both stress controls. A curve-table action updates baseline only, preserving selected dropout. Analytical estimates do not run Monte Carlo for every cell.

Simulation seed

42

Integer · 0–4,294,967,295

Simulation replicates

10,000

Integer · 1–100,000

Run simulation

Not yet run · No simulation estimate available. Power is conditional on assumptions, not the probability that a drug works.

Download current scenario JSON · unavailable

Run a current selected-scenario simulation before exporting a complete record.

Route / · Ready only with matching valid grid point and available service adapter. Capture frozen design, scenario, method/hash, seed 42 and 10,000 replicates on Run. Settings edits invalidate results. Backend progress/error live region; retry same captured context. No estimate until service returns.

Better Baseline

LIVE / SERVICE BINDINGS · 09 / INVALID · Route /

This scenario cannot be evaluated

INVALID · Analytic power: null — not 0%

Inherited ARR 0.12 (12pp); stress control risk 10% (0.10), dropout 0% (0.00). Treatment event risk −0.02 is outside the model domain. Choose a valid scenario or revise planning and recalculate.

Enrollment remains locked: ‹N_C› / ‹N_T›. Expected retained counts may be shown only when retention inputs are valid.

Curve: gap at invalid baseline · Grid: null ×

Risk 10% (0.10) / dropout 0% (0.00) / Power null / Invalid true
Select invalid cell: disabled · Simulation: blocked · Δ pp unavailable
Target 80% · Cohen’s-h / normal approximation
Hash ‹matching hash› · version ‹method version› · warnings ‹domain diagnostics›

Run simulation · blocked: invalid scenario

Download current scenario JSON · unavailable

02 / STRESS SCENARIO

Change assumptions, not enrollment

Control endpoint risk · 10%



0.10
fraction (0.10 = 10%)

Symmetric dropout · 0%



0.00
fraction (0.10 = 10%)

Inherited ARR 0.12 · stress risk 0.10 implies treatment risk −0.02. Choose a valid risk or revise planning and recalculate.

Evaluate Stress Scenario

Load grid response from backend

Risk 1–99% and dropout 0–50%; 1pp steps, numeric fractions. Edited assumptions debounce 350ms and may be evaluated explicitly. Invalid grid cells are not selectable.

↑ Revise planning assumptions

Invalid cells cannot be selected for simulation. Provide model-domain reason from diagnostics; never bridge null regions in trace or target boundary. Announce invalid status explicitly, not through color alone.

10 / SELECTED-SCENARIO SIMULATION

One scenario. Reproducible execution.

EXECUTION SETTINGS

Seed (integer)

42

0–4,294,967,295 · default 42

Replicates (integer)

10,000

1–100,000 · default 10,000

Requires a valid current LIVE design, selected scenario and available live adapter. Enrollment remains frozen.

RUNNING · Monte Carlo simulation in progress...

Seed 42 · Requested replicates 10,000
Captured control risk 10% · symmetric dropout 0%
Locked control / treatment ‹N_C› / ‹N_T›
Request inputs hash ‹captured hash›

Indeterminate progress: do not invent a completion percentage.

Running simulation...

Download current scenario JSON · unavailable

Alternate state · FAILED

The simulation service could not complete this request. Your inputs and enrollment lock are unchanged. Retry when the adapter is available.

Retry simulation

No current simulation result yet. Preserve entered settings and frozen enrollment. Disable duplicate submission; announce running and completion using a polite live region.

10 / FUNCTION MAP · Polite live region announces running/completion. Duplicate submission disabled. On assumptions, selection, seed or replicate edit, invalidate capture and discard late responses. Prior captured output may remain only as stale. Retry preserves inputs and frozen enrollment.

Better Baseline

Trial planning / assumption stress testing

LIVE

11 / SELECTED-SCENARIO SIMULATION

Inspect the result, then preserve the record

ILLUSTRATIVE SERVICE-BINDING SPECIFICATION · Current-result UI, not an actual run. ‹Fields› bind returned outputs; no verified numbers or identifiers are fabricated.

EXECUTION SETTINGS

Seed (integer)

42

0–4,294,967,295 · default 42

Replicates (integer)

10,000

1–100,000 · default 10,000

Requires a valid current LIVE design, selected scenario and available live adapter. Enrollment remains frozen.

CURRENT · Design, selection, curve, grid and simulation match

Risk 10% · dropout 0% · RR 0.75 · two-sided α 5% · target 80%
Frozen enrollment ‹N_C› / ‹N_T› · seed 42 · requested replicates 10,000

Selected-scenario Monte Carlo

Two-sided significance fraction

‹two-sided fraction›

MCSE ‹two-sided MCSE› percentage points

Favorable-direction significance fraction

‹favorable fraction›

MCSE ‹favorable MCSE› percentage points

Invalid-test fraction

‹invalid fraction›

‹Undefined tests› / ALL actual replicates. Undefined tests are nonrejections, never omitted.

Actual replicates ‹returned count› · seed ‹returned seed›
Both significance fractions use ALL actual replicates. Favorable = significant in the lower-adverse-risk treatment direction.

Enrollment is not retention

Frozen enrolled before losses

Control ‹N_C› / Treatment ‹N_T›

Floored expected evaluable · analytic

Control ‹expected_C› / Treatment ‹expected_T›

Mean sampled actual retained · simulation

Control ‹mean retained_C› / Treatment ‹mean retained_T›

Mean simulated events

Control ‹mean events_C› / Treatment ‹mean events_T›

Inflation accounts for expected loss; it does not guarantee retained counts. Monte Carlo samples actual independent retention.

Methods & execution provenance

Method ‹returned method; expected pooled two-proportion z test›
Model version ‹returned version› · RNG ‹returned generator›
Software versions ‹returned environment› · Timestamp ‹returned UTC›
Inputs hash ‹returned hash› · Result identifier ‹returned result ID›

Returned field provenance: source, description and source URL where available for each output. Verification/retrieval metadata preserved; no clinical source invented.

Simulation warnings ‹returned warnings, or explicit none returned›
Invalid tests are nonrejections; analytic and simulation methods are different approximations.

A complete, matching scenario record

Available only after every current input, design, grid/curve and selected simulation matches the displayed seed and replicate count.

Download current scenario JSON

Record contains assumptions, fixed enrollment, stress, analytic design/grid/curve, simulation and separate MCSEs, warnings/provenance, seed/replicates, methods/models/software/RNG, timestamps and hashes.

MCSE measures simulation sampling error, not uncertainty in the assumed rates or biological effect. More replicates improve estimate precision, not a real trial's success probability.

11 / FUNCTION MAP · CURRENT is a UI state specification. Enable JSON only on complete, matching design/grid/curve/simulation/settings. Download the complete artifact, not the visible summary.

12 / SELECTED-SCENARIO SIMULATION

Changed settings. Previous result is not current.

STALE · Replicate count changed to 20,000

Previous simulation: seed 42 / 10,000 replicates / ‹old inputs hash›. Keep prior estimates only under this stale label; do not use them in current export.

EXECUTION SETTINGS

Seed (integer)

42

0–4,294,967,295 · default 42

Replicates (integer)

20,000

1–100,000 · default 10,000

Requires a valid current LIVE design, selected scenario and available live adapter. Enrollment remains frozen.

Simulation seed: 42
Simulation replicates: 20,000
Current selected scenario: ‹risk› / ‹dropout› · lock ‹N_C› / ‹N_T›

Run simulation

Download current scenario JSON · unavailable

Unavailable: displayed replicate count does not match captured simulation. Run a matching simulation; ensure design and analytic data are current.

INVALID INPUT · Seed −1 / replicates 100,001

Seed must be an integer between 0 and 4,294,967,295. Replicates must be an integer between 1 and 100,000. Correct settings before running.

Run simulation · blocked by invalid settings

12 / FUNCTION MAP · Scenario, assumptions, seed or replicate edits invalidate results. JSON stays unavailable until design, grid, curve, simulation and settings match. Never mix stale output with new inputs or fabricate missing fields.

Better Baseline

LIVE · Calculation service

The service did not finish. Your work is safe.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)
20
Enter 20 for 20%. Risk in eligible control participants over the specified follow-up, not prevalence. Valid: >0 and <100.

Hypothetical effect / RR or ARR
RR · ratio | ARR · pp
Switch RR / ARR with arrow keys. Hypothetical effect, not efficacy evidence.

Relative risk (ratio)
0.75
Decimal ratio; strictly >0 and <1.

Dropout in each arm (%)
0
One shared value for both arms. Valid: 0 to <100.

Target power (%)
80
>0 and <100; must exceed alpha.

Two-sided alpha (%)
5
>0 and <100.

Retry calculation

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

Failed · no new lock

The calculation service is unavailable. Your entered assumptions are preserved. Check the local backend connection and retry.

Unavailable · not zero

No current power returned.
Existing lock, if any: retained as stale.
Stress evaluation/grid: unavailable until service recovers.

Source: synthetic preset
Follow-up: {preset.follow_up_label}
No clinical source asserted.

Retry stress evaluation · when valid

LIVE ≠ clinical evidence

Fresh numerical results require the calculation service. A local backend can work without a registry or internet connection. Mode is externally configured; no mode toggle.

Load grid response from backend

Grid retry requires a valid current design and an available adapter. Failure leaves stale outputs explicitly labeled, never substituted with zero.

13 / INTERACTION · RECOVERY

Show an understandable service error, preserve inputs and prior lock, and announce failure with role=alert. Offer Retry calculation, Evaluate Stress Scenario, and “Load grid response from backend” retries in their respective failed regions. Never promote fixture or late results to current.

Better Baseline

DEMO · Illustrative only

Explore the fixture. No backend execution.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

01 / Planning assumptions

Equal allocation · binary adverse outcome · two-sided superiority

Control endpoint risk (%)
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Decimal ratio; strictly >0 and <1.

Dropout in each arm (%)
0
One shared value for both arms. Valid: 0 to <100.

Target power (%)
80
>0 and <100; must exceed alpha.

Two-sided alpha (%)
5
>0 and <100.

Live calculation unavailable · DEMO

Successful calculation sets a new enrollment lock. Editing never silently replaces an existing lock.

02 / Enrollment lock

Illustrative fixture · not live

Baseline collapse hand-authored fixture. Not newly calculated, and not a verified simulation result.

DEMO fixture · enrollment held fixed
Control enrolled: 903
Treatment enrolled: 903
Stressed power: approximately 47%
Selected risk: 10% · dropout: 0%

Source: synthetic preset
Follow-up: {preset.follow_up_label}
No clinical source asserted.

Run simulation · unavailable in DEMO

Complete scenario JSON unavailable

DEMO uses fixtures/client calculations, no backend execution. Monte Carlo cannot run; complete export requires matching live simulation. Analytic grid CSV remains a separate download.

Download analytic grid CSV

Complete scenario JSON · unavailable

14 / INTERACTION · DEMO RESTRICTIONS

Synthetic inputs can be used in genuine LIVE calculations; synthetic input is not demo output. Mode is external, with no toggle. Other supplied fixtures only: Attrition shock 951/arm, ≈50% stressed power; Stable design 951/arm, ≈78%. All illustrative; never invent simulation output.

Better Baseline

LIVE · Calculation service

Validate a record, not a restored project.

Plan a hypothetical trial. Lock enrollment. Stress-test assumptions.

SYNTHETIC PRESET / 3 EXAMPLES

Selected · Baseline collapse
20% → 10% risk · 0% → 0% dropout
ID/version: {baseline_collapse.identity}
Preset hash: {baseline_collapse.hash}

Attrition shock
20% → 20% risk · 5% → 25% dropout
ID/version: {attrition_shock.identity}
Preset hash: {attrition_shock.hash}

Stable design
20% → 19% risk · 5% → 6% dropout
ID/version: {stable_design.identity}
Preset hash: {stable_design.hash}

Lower control event risk, unchanged hypothetical effect. Synthetic assumptions; not clinical evidence.

All presets: RR 0.75 · target 80% · two-sided α 5% · seed 42 · 10,000 replicates

Reset to Preset Defaults

Change or reset restores preset assumptions, seed and replicates; invalidates pending work and captured results.

Scenario record

Export a complete, current record only when all inputs and results match. Validate a previously downloaded file without changing this workspace.

Download current scenario JSON

Export unavailable

No complete matching result: calculate a current design, refresh analytic curve/grid, and run the selected live simulation with the displayed seed and replicate count. Never combine stale results with new inputs.

Record includes assumptions, frozen enrollment, stress, analytic design/curve/grid, simulation outputs, separate MCSEs, warnings, field provenance, seed/replicates, method/model/software versions, RNG, timestamps and hashes.

Reopen and validate scenario JSON

Choose JSON file

Accepts .json · validation only; not patient data

File validation feedback

Valid JSON · detached artifact

Schema: {validated.schema_version}
Export time: {validated.export_time}
Seed: {validated.seed}
Replicates: {validated.replicates}

The artifact matches the schema. Your current assumptions and results are unchanged.

Invalid file · alternate feedback

Schema validation failed: required fields are missing or malformed.
{validation.error_path}: {validation.error_message}
Choose a complete scenario JSON and retry; workspace unchanged.

Choose another JSON file

These are alternate success/error states, shown together for implementation review. Display only the applicable feedback in the workspace; returned metadata uses service/file bindings.

15 / INTERACTION · DETACHED VALIDATION

Use a labeled file input accepting JSON. Announce success via status and errors via alert; keep keyboard focus on the validator. Reopening never restores assumptions, replaces the lock, or reruns calculations. Schema validity does not independently prove statistical correctness. Reproduction requires the recorded software environment.

Trial Planning Workspace / Expanded audit trail

Assumptions, methods & provenance

STALE / UNHASHED · Manual planning edits do not have a current result hash. Last design hash ‹previous service hash› remains stale.

Tracked field	Value / unit	Source & description	Source / retrieval / verification
Control endpoint risk	22% · manual edit	Manual-user · hypothetical control event probability	No clinical URL · unhashed
Effect model / value	RR / 0.75	Synthetic preset · hypothetical risk ratio	‹versioned preset identity›
Target / alpha	80% / 5%	Synthetic preset · planning thresholds	‹inputs hash› · no clinical source
Planned dropout	0% each arm	Synthetic preset · symmetric independent losses	‹available source metadata›
Stress control risk / dropout	0.10 / 0.00 fractions	Manual-user · selected stress assumptions	No clinical source · selection hash pending
Seed / requested replicates	42 / 10,000 integers	Synthetic preset defaults · execution settings	Invalidated by edits
Design / analytic / simulation outputs	‹returned fields›	Calculation-service field provenance	‹URL if available› · ‹retrieval / verification metadata›

Tracked fields show source, description and available URL. Preset selection/reset restores synthetic provenance, assumptions and simulation defaults and invalidates captured/pending work. Edited fields acquire manual-user provenance. Preserve supplied generator, scenario identity, follow-up label, model/method versions and warnings.

Available source & result metadata

Scenario identity: ‹returned scenario identity›
Follow-up: ‹source follow-up label› · descriptive context only
Generator: ‹supplied generator›
Source URL: ‹available source URL, otherwise not supplied›

Method / model: ‹returned versions›
Result identity / hash: stale · ‹previous identifiers›
Retrieval & verification: ‹supplied metadata›
Current input hash: unavailable after manual edits

What the model means

Baseline risk

A specified binary event in eligible control participants over a stated follow-up period; not general disease prevalence.

Hypothetical treatment risk

Implied by RR or ARR, not evidence of efficacy.

Supported model

Two arms, equal allocation, binary adverse-outcome superiority, two-sided test, independent symmetric dropout.

Analytic method

Cohen’s-h / normal approximation with floored expected retained counts. Conditional power depends on assumptions and method.

Simulation method

Sample actual retained counts; pooled two-proportion z test. This differs from the analytic approximation.

Enrollment and retention

Enrolled counts are before losses. Required evaluable counts, floored expected retained counts and sampled actual retention are different quantities. Inflation for expected losses never guarantees retained counts.

What the model does not claim

MCSE

Simulation sampling error only; not uncertainty in assumed event rates or biological effect.

Replicate precision

More replicates improve simulation precision, not variability or success of a real trial.

Two different rejections

Two-sided and favorable-direction rejection differ; under a symmetric null they are approximately α and α/2.

Target-power envelope

Hypothetical assumptions meeting the chosen threshold; not a clinical or regulatory safe zone.

No certification or prediction

Does not predict drug efficacy, FDA approval, trial or commercial success, or cost savings. Does not provide clinical/regulatory design certification.

WARNINGS & REPRODUCIBILITY · ‹Returned design, grid and simulation diagnostics›. Source metadata does not establish clinical truth. Schema-valid JSON does not prove statistical correctness; reproduction requires the recorded software environment.

In-workspace disclosures use labelled disclosure buttons with expanded/collapsed state. Tables have captions and scoped headers; charts have numerical alternatives. Minimum 44px targets, visible keyboard focus, no status conveyed by color alone. Announce evaluation, invalidity, errors and export availability. No mode toggle or extra routes.

STATE 16 / Same route /. Expanded Methods, warnings & field provenance disclosure. Collapsing this region returns to the working surface; it does not navigate. Manual fields stay unhashed until recalculation. Returned metadata is displayed only when supplied.

Collapse methods & provenance –